MATH 3312, SPRING 2026: HOMEWORK 8

(1) Suppose that R is an integral domain. Show that every (non-zero) prime element $r \in R$ is irreducible.

Definition 1. The **sum** $I_1 + I_2 \leq R$ of two ideals I_1 and I_2 is the ideal consisting of elements of the form $x + y$ with $x \in I_1$ and $y \in I_2$.

Definition 2. Two ideals $I_1, I_2 \leq R$ in a commutative ring R are **relatively prime** if $I_1 + I_2 = R$.

(2) Let R be a PID and suppose that $q_1, q_2 \in R$ are two irreducible elements generating *distinct* ideals. Show that, for any $n_1, n_2 \geq 1$, the ideals $(q_1^{n_1}), (q_2^{n_2})$ are relatively prime.

Definition 3. If $I_1, I_2 \leq R$ are two ideals, then their **product** $I_1 I_2$ is the ideal generated by elemens of the form xy with $x \in I_1$ and $y \in I_2$.

(3) (Chinese Remainder Theorem: General version) Suppose that $I_1, I_2 \leq R$ are two relatively prime ideals.

(a) Show that the ring homomorphism

$$R \xrightarrow{r \mapsto (r + I_1, r + I_2)} R/I_1 \times R/I_2$$

is surjective.

(b) Show that the kernel of the map in (a) is $I_1 I_2 = I_1 \cap I_2$.

(c) Show more generally that if $I_1, \ldots, I_m \leq R$ are ideals such that, for any $r \neq s$, I_r and I_s are relatively prime, then we have an isomorphism[1]

$$R/(I_1 \cdots I_m) \xrightarrow{\simeq} R/I_1 \times \cdots \times R/I_m.$$

(d) Suppose that R is a PID with $q_1, \ldots, q_m \in R$ irreducible elements generating *distinct* ideals. Show that, for any $n_1, \ldots, n_m \geq 1$, we have

$$R/(q_1^{n_1} \cdots q_m^{n_m}) \xrightarrow{\simeq} R/(q_1^{n_1}) \times \cdots \times R/(q_m^{n_m}).$$

Hint: For (a), use the fact that we can write $1 = j_1 + j_2$ with $j_1 \in I_1$ and $j_2 \in I_2$. For (b), the main thing is to show the equality $I_1 I_2 = I_1 \cap I_2$, and this also follows from the same observation.

Definition 4. Suppose that $A \in M_n(k)$ is conjugate to a matrix in rational canonical form corresponding to polynomials $f_1(x) \mid \cdots \mid f_m(x)$. Then the **minimal polynomial** of A, $\min_A(x)$ is the polynomial $f_m(x)$, while the **characteristic polynomial** of A, $\mathrm{ch}_A(x)$ is the polynomial $f_1(x) \cdots f_m(x)$.

(4) Show that the following are equivalent for a matrix $A \in M_n(k)$ and $\lambda \in k$:

(a) λ is a zero for $\min_A(x)$.

(b) λ is a zero for $\mathrm{ch}_A(x)$.

(c) The k-linear transformation $k^n \xrightarrow{v \mapsto T_A(v) - \lambda \cdot v} k^n$ is not an isomorphism.

(d) There exists a non-zero vector $v \in k^n$ such that $Av = \lambda v$.

Hint: Think about the k-pair V_A as a $k[x]$-module: What does multiplication by $x - \lambda$ do in terms of the expression for V_A given by the rational canonical form?

Definition 5. A $\lambda \in k$ satisfying the equivalent conditions of the previous problem is called an **eigenvalue** for A.

[1] The product of multiple ideals is defined iteratively.

If $p(x) = a_0 + \cdots + a_{d-1}x^{d-1} + a_d x^d \in k[x]$ is a polynomial and $A \in M_n(k)$, we set

$$p(A) = a_0 I_n + \cdots + a_{d-1}A^{d-1} + a_d A^d \in M_n(k).$$

(5) Show that the following are equivalent:
  (a) $p(A) = 0 \in M_n(k)$.
  (b) $\min_A(x) \mid p(x)$.

Conclude that we always have $\mathrm{ch}_A(A) = 0$: this is the **Cayley-Hamilton theorem**.

This explains the term minimal polynomial*: It is the smallest polynomial that kills A.*

Hint: You want to use the fact that

$$V_A \simeq k[x]/(f_1(x)) \times \cdots \times k[x]/(f_m(x))$$

and note that multiplication by A on the left corresponds to multiplication by x on the right.

(6) Let $M \leq \mathbb{Z}[i]^3$ be the $\mathbb{Z}[i]$-submodule generated by

$$(1+i, 1-i, 2), (2+i, 2-i, 5), (1+2i, 1-2i, 5).$$

Find the invariant factor and elementary divisor decomposition for $\mathbb{Z}[i]^3/M$.

Hint: You might want to show that $1 \pm i$, $2 \pm i$, $1 \pm 2i$ are all prime elements, though they only generate three distinct ideals among them.

(7) A polynomial $f(x) \in \mathbb{Z}[x]$ is **primitive** if its coefficients have gcd 1. For example, $2x + 3$ is primitive, but $2x + 4$ is not.
  (a) Show that $f(x) \in \mathbb{Z}[x]$ is primitive if and only if, for every prime p, the image of $f(x)$ in $\mathbb{F}_p[x]$ is non-zero.
  (b) Use this to show that the product of two primitive polynomials is once again primitive.
  Hint: $\mathbb{F}_p[x]$ is an integral domain.
  (c) Suppose that $f(x) \in \mathbb{Q}[x]$ is a non-zero polynomial. Show that there exists a positive rational number $r \in \mathbb{Q}$ such that $r \cdot f(x)$ is a primitive polynomial in $\mathbb{Z}[x]$.
  (d) Suppose that $f(x) \in \mathbb{Z}[x]$ is a polynomial with leading coefficient 1, and that $h(x), g(x) \in \mathbb{Q}[x]$ are also polynomials with leading coefficient 1 such that $f(x) = h(x)g(x)$. Show that $h(x), g(x) \in \mathbb{Z}[x]$.
  Hint: Apply the previous part to $h(x)$ and $g(x)$ and see what that tells you.

The fourth statement is usually called Gauss's lemma.